

Explainable Diabetic Retinopathy Detection using Deep Learning

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Abstract - Diabetic Retinopathy (DR) is a critical complication of diabetes that leads to progressive vision impairment and, if left untreated, permanent blindness. With the increasing prevalence of diabetes worldwide, early and accurate detection of DR has become essential for timely intervention. Traditional manual screening methods are time-consuming, prone to human error and require skilled ophthalmologists, which limits their scalability. Recent advances in deep learning have enabled the development of automated systems that can detect and classify DR from retinal fundus images with high accuracy. However, most of these models function as "black boxes" and lack transparency, making their predictions difficult to interpret in clinical settings. This paper presents a comprehensive and explainable DR detection pipeline using Convolutional Neural Network (CNN) architectures such as VGG16, Xception, ResNet50 and Sequential CNN. A publicly available real-world dataset of labeled fundus images is used for training and evaluation. Through comparative analysis, the performance of each model is assessed based on key metrics including accuracy, precision, recall and F1-score. The ultimate goal is to aid ophthalmologists with a reliable, accurate and interpretable diagnostic tool, while also laying the groundwork for the future incorporation of explainable AI techniques such as Grad-CAM and synthetic image generation using GANs.

Key Words - Diabetic Retinopathy, Deep Learning, CNN, Medical Imaging, Transfer Learning

1. INTRODUCTION

By 2040, approximately 600 million people are predicted to have diabetes and one third are expected to have diabetic retinopathy [1]. Diabetes reduces life expectancy by five to 10 years [2]. Diabetic retinopathy is the most common microvascular complication in diabetes [3]. It affects people with diabetes and is the result of prolonged high blood glucose levels that damage retinal blood vessels. Early detection can prevent vision loss, but access to ophthalmologists is often limited. Retinal imaging is the most widely used method due to its high sensitivity in detecting retinopathy [3].

As the number of patients with diabetes increases, the number of retinal images produced by screening programs will also increase, placing a large burden on medical experts and healthcare systems. This could be alleviated with an automated system. Studies show that automated systems based on deep learning neural networks can achieve high sensitivity and specificity in detecting referable diabetic retinopathy. Other referable eye complications such as diabetic macular edema and glaucoma have also been explored [3].

Artificial Intelligence (AI), especially deep learning, has become a powerful ally in healthcare. Convolutional Neural Networks (CNNs) can automatically classify retinal images and detect DR stages. Despite their potential, these models often lack transparency in their predictions, which can be a barrier to clinical trust. This research proposes a deep learning pipeline for DR detection using various CNN architectures. We emphasize accuracy, robustness and the foundation for future interpretability enhancements.

2. DATASET AND PREPROCESSING

We used a fundus image dataset containing 3630 labeled images across five DR classes: No DR, Mild, Moderate, Severe and Proliferative DR.

2.1 Preprocessing

- All images were resized to 224 × 224 pixels.
- Pixel intensities were normalized to [0, 1].
- Data augmentation (rotation, flip, brightness) was used to balance classes and reduce overfitting.
- One-hot encoding was used for the labels.

2.1 Dataset Description

We used the publicly available APTOS 2019 Blindness Detection dataset collected by Aravind Eye Hospital, India [4]. It originally had 3630 Images, but after dataset balancing it now contains 4530 labeled images rated for DR severity on a scale of 0 to 4. Distribution: 1016 (No DR), 840 (Mild), 999 (Moderate), 790 (Severe), 885 (Proliferative DR). We split the

dataset into training (80%), validation (10%) and testing (10%) subsets.

2.2 Description of Retinal Fundus Images

- **Normal Retina:** Clear optic disc, distinct vessels, no hemorrhages or exudates.
- **Mild DR:** Microaneurysms visible as small red dots.
- **Moderate DR:** Scattered exudates and hemorrhages.
- **Severe DR:** Widespread hemorrhages and exudates.
- **Proliferative DR:** Signs of neovascularization.

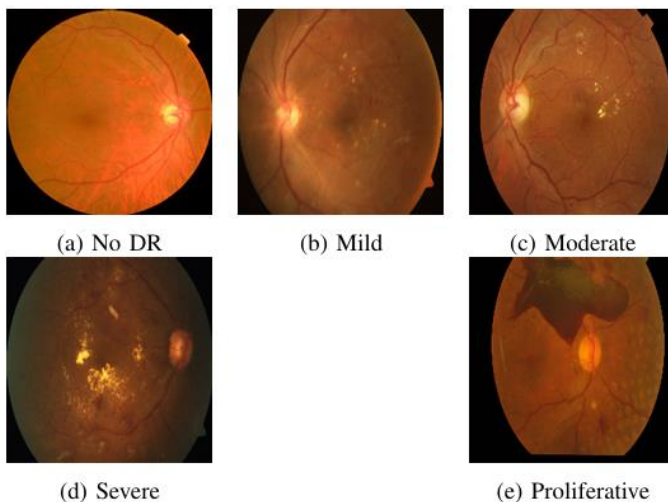


Fig-1: Sample fundus images for each DR class

3. MODEL ARCHITECTURES

To build a robust diabetic retinopathy (DR) detection system, four pre-trained convolutional neural network (CNN) models were adapted and evaluated: Xception, ResNet50, VGG16 and InceptionV3. Each of these architectures offers distinct advantages in terms of depth, computational efficiency and feature representation capabilities, which are critical for the complex task of retinal image analysis [5], [6].

3.1 Xception

The Xception architecture replaces standard convolutional layers with depthwise separable convolutions, enabling independent spatial and channel-wise filtering. This design significantly reduces model complexity without compromising performance [7]. Structurally, Xception is divided into three primary components: the entry flow, middle flow and exit flow, which progressively learn increasingly abstract representations of input images. Its architecture is well-suited for identifying minute retinal abnormalities such as microaneurysms and hemorrhages. In this project, the Xception model consistently achieved the

highest classification accuracy among the compared networks, indicating its effectiveness in fine-grained medical image classification tasks [7].

3.2 ResNet 50

ResNet50 is a 50-layer deep CNN that incorporates residual connections, or skip connections, to enable stable training of deep models. These connections allow gradients to propagate more efficiently during backpropagation, thereby addressing the vanishing gradient problem common in deep architectures [6]. Though ResNet50 showed moderate training accuracy in this study, its validation performance was promising, indicating potential for further optimization. Its depth and design make it suitable for learning hierarchical and complex visual patterns in retinal fundus images.

3.3 VGG16

VGG16 is a well-known CNN model characterized by its uniform use of small (3×3) convolutional kernels throughout the network. Comprising 13 convolutional layers and 3 fully connected layers, the architecture is both deep and straightforward. Despite its relatively large parameter count, VGG16 has proven to be highly effective in transfer learning scenarios [5]. In this research, the model exhibited a strong balance between training and validation accuracy, demonstrating excellent generalization capability for DR classification. Its predictable structure and stability make it an ideal choice for medical imaging problems.

3.4 InceptionV3

The InceptionV3 model employs a sophisticated design that utilizes inception modules, which apply multiple filter sizes in parallel to capture information at different scales. Additionally, it incorporates techniques such as factorized convolutions and dimensionality reduction to enhance computational efficiency while preserving learning capacity. This multi-path approach enables the model to extract a rich set of features from high resolution retinal images. In our experimentation, InceptionV3 demonstrated consistent performance, particularly in distinguishing between intermediate DR stages, due to its ability to detect both coarse and fine patterns [6].

3.5 Transfer Learning Setup

The transfer learning strategy included the following steps to fine-tune the models for DR classification:

- Applied ImageNet pre-trained models: Xception, ResNet50, VGG16 and InceptionV3 [6], [8].
- Removed original top layers using include_top=False.

- Input dimensions: Xception at 299×299, others at 224× 224.
- Froze base layers to preserve pre-trained weights.
- Added a custom classification block:– Global average pooling– Dropout layer (rate = 0.5)– Dense layer with softmax activation for 5 DR classes
- Loss function: categorical cross-entropy
- Optimizer: Adam (learning rate = 0.0001)
- Training setup: 25–30 epochs, batch size of 32
- Training callbacks: EarlyStopping, learning rate scheduler
- Augmentation: random flip, rotation, zoom and brightness shift.

Model	Params	Input Size	Key Features
hline Xception	~22M	299×299	Separable convolutions
ResNet50	~25M	224×224	Residual connections
VGG16	~138M	224×224	Uniform 3×3 kernels
InceptionV3	~23M	224×224	Multi-scale modules

Table -1: Comparison of CNN Models

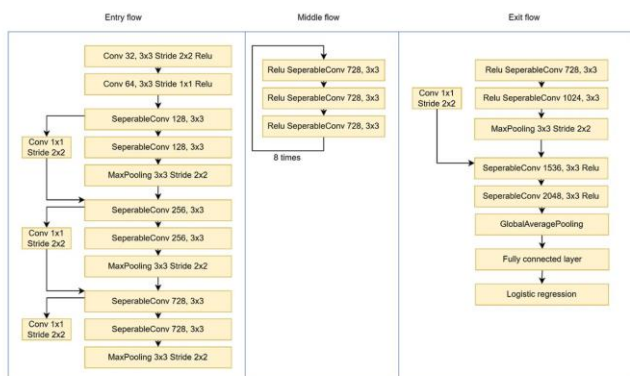


Fig-2: Xception Architecture

4. TRAINING AND EVALUATION

4.1 Training Strategy

The training pipeline involved fine-tuning several deep convolutional neural networks to classify retinal fundus images into five diabetic retinopathy (DR) stages: No DR, Mild, Moderate, Severe and Proliferative DR. The dataset used for this task comprised 4,535 annotated fundus images.

To enhance model performance and reduce overfitting, the following preprocessing strategies were employed:

- Image Resizing: All images were resized to 224 × 224 pixels to conform to the input requirements of architectures such as MobileNetV2 and Xception.

- Normalization: Pixel values were scaled to the range [0, 1] to improve optimization consistency and convergence speed [5].
- Data Augmentation: Techniques such as random rotation, flipping, zooming and brightness adjustments were applied to improve generalization and prevent overfitting [6].

4.2 Evaluation Metrics

The models were evaluated using standard classification metrics:

- **Accuracy:** Measures the overall correctness of predictions across all classes.
- **Precision:** Indicates the ratio of correctly predicted positive observations to total predicted positives, helping minimize false positives.
- **Recall (Sensitivity):** Represents the proportion of correctly predicted positives among all actual positives, which is vital in medical diagnostics.
- **F1-Score:** Harmonic mean of precision and recall, particularly effective when dealing with class imbalance [7].
- **AUC-ROC:** Evaluates the model's capability to distinguish between classes and is considered robust across imbalanced datasets.

4.3 Model Performance

The performance of various CNN models is summarized in Table II. Notably, the Xception model achieved the best balance between accuracy and generalization.

Model	Training Accuracy	Validation Accuracy
Xception	96.72%	91.88%
VGG16	92.67%	92.05%
Sequential CNN	98.00%	90.84%
ResNet50	72.05%	75.30%

Table - 2: Model Performance on DR Classification

These findings align with prior research, such as Sahlsten et al., who demonstrated the effectiveness of Inception-based and ensemble models for DR grading [5] and other studies advocating preprocessing and augmentation as critical steps in training robust models [5].

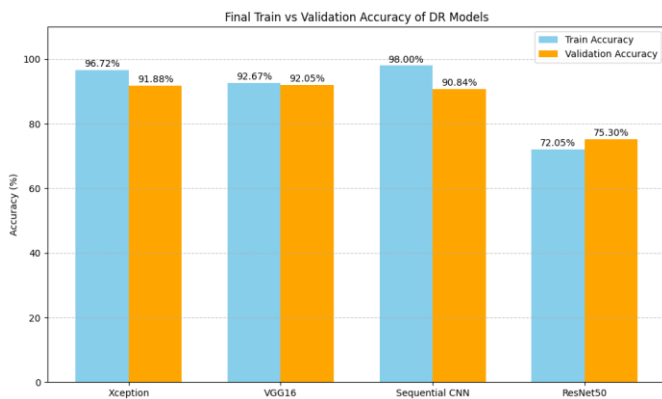


Fig-3 : Training and validation accuracy over epochs

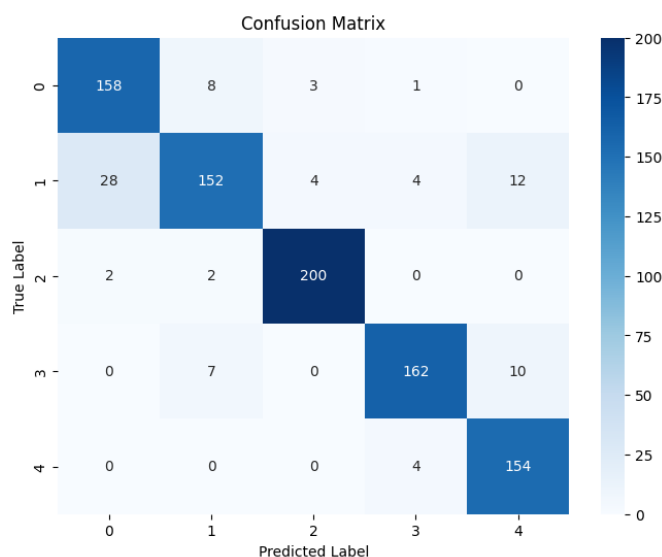


Fig-4 : Confusion Matrix for Xception model

5. RESULTS AND ANALYSIS

Xception outperformed other models in overall accuracy. VGG16 showed stability and good generalization. ResNet50 underperformed possibly due to shallow fine-tuning or insufficient feature extraction.

6. CONCLUSION AND FUTURE SCOPE

Deep learning, particularly CNN architectures like Xception and VGG16, show great promise for DR detection. Preprocessing and data augmentation play a critical role in performance.

6.1 Future Scope

- **Explainability:** Incorporate Grad-CAM and LIME.
- **GANs:** Use Patho-GAN for synthetic image generation.

- **Web Integration:** Deploy model in clinical workflows.

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